CSE370 : Database Systems
Project Report
Project Title : Smart Inventory Database for Medical Supplies

Group No : _8_, CSE370 Lab Section : _05_, Summer 2025		
ID	**Name**	**Contribution**
22201484	Sabah Maryam	Dashboard, Transfer, Frontend Content management (Header)
21301563	Shezan Mahmud	Inventory Management, Document Manager, Shortage & Expiry Alerts
21301655	Izaz Mahmud Evan	OpsCenter, Export Center, Frontend Content management (Footer)

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Introduction

Project Overview:

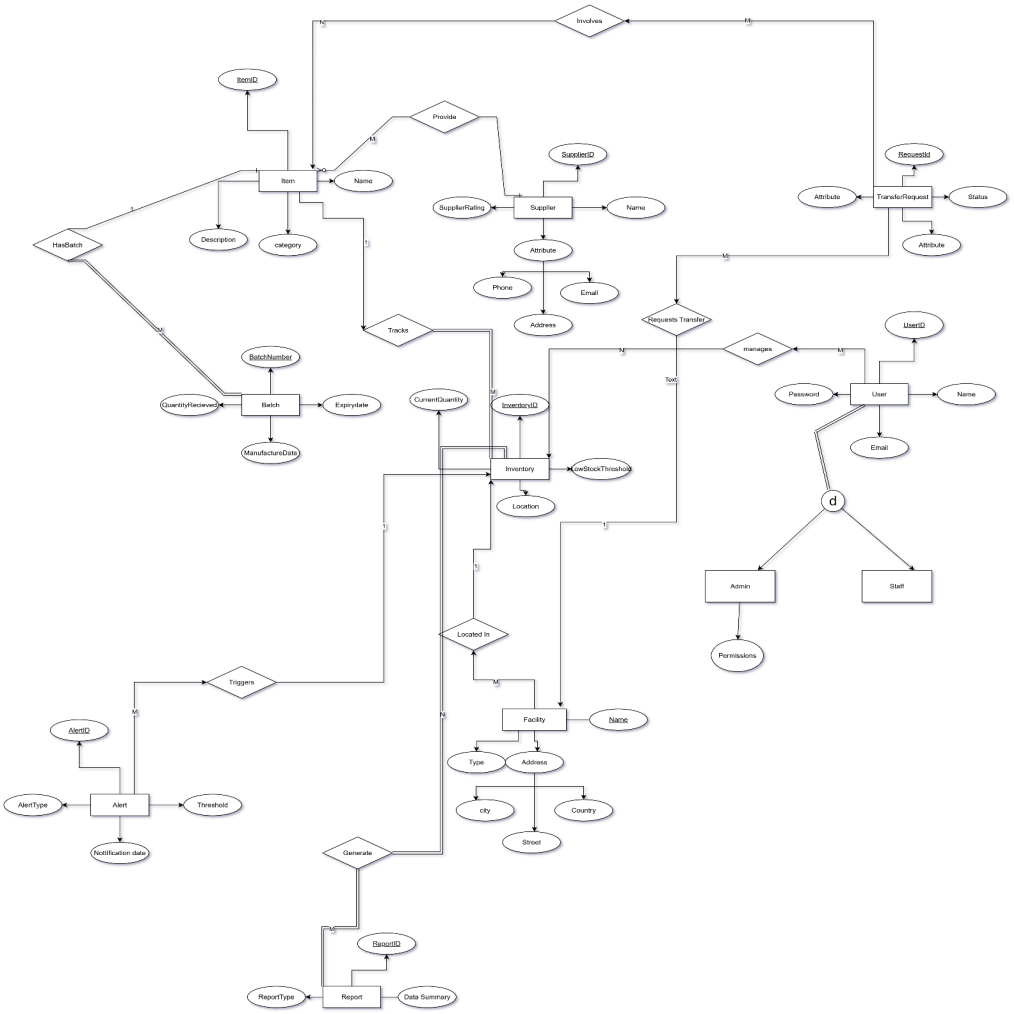
A Smart Inventory Database for Medical Supplies is a web-based platform designed to enhance inventory management for hospitals/clinics in Bangladesh, while seamlessly connecting them with medical suppliers. The system provides real-time tracking of medical supplies, automated alerts for low stock and upcoming expirations, and supports stock addition, consumption, and transfer within and across healthcare facilities.Crucially, when hospitals or clinics face inventory shortages, suppliers can instantly view these demands via their dedicated supplier dashboard in the system. This visibility enables suppliers to proactively deliver the required supplies according to the healthcare providers' needs, speeding up replenishment and minimizing stock-outs.

The platform is equipped with role-based access control, detailed reporting, supplier and batch management, and facilitates resource sharing across locations. By addressing local challenges in inventory accuracy, shortage reduction, and waste minimization, this integrated solution optimizes resource utilization and strengthens collaboration between healthcare providers and suppliers—ultimately improving patient care and operational efficiency across the healthcare

Project Features

ID, Name	Features [3 per member]	
22201484, Sabah Maryam	Ft 1	Dashboard
	Ft 2	Transfer
	Ft 3	Frontend Content management(Header)
21301563, Shezan Mahmud	Ft 1	Inventory Management
	Ft 2	Document Manager
	Ft 3	Shortage & Expiry Alerts
21301655, Izaz Mahmud Evan	Ft 1	OpsCenter
	Ft 2	Export Center
	Ft 3	Frontend Content management(Footer)

ER DIAGRAM OF
Smart Inventory Database for Medical Supplies



SCHEMA DIAGRAM

SMART INVENTORY DATABASE MANAGEMENT

SUPPILER

SUPPLIERID	NAME	SUPPLYRATING	PHONE	EMAIL	ADDRESS

TRANSFERREQUEST

REQUESTID	STATUS	ADD	USE

ITEM

ITEMID	STOCK	CATEGORY	DESCRIPTION

BATCH

BATCHNO	QUANTITYRECIEVED	EXPIRYDATE	UNIT

USER

USERID	NAME	EMAIL	PASSWORD

REPORT

REPORTID	REPORTTYPE	DATASUMMARY

INVENTORY

INVETORYID	LOWSTOCKTHRESHOLD	LOCATION	STOCK

ALERT

ALERTID	ALERTTYPE	THRESHOLD	NOTIFICATION DATA

FACILITY

FNAME	TYPE	COUNTRY	CITY	STREET

STAFF

USERID

ADMIN

USERID	PASSWORD

REQUESTTRANSFER

REQUESTID	FNAME

HASBATCH

BATCHNO	ITEMID

provide

ITEMID	SUPPLIERID

MANAGES

INVENTORYID	USERID

triggers

ALERTID	INVENTORYID

LOCATEDIN

FNAME	INVENTORYID

Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

a . Answer : Yes,it decomposed in 1NF because there are no multivalued or composite attributes.

The relations of this database's schema only have primary key and single attributes.

b . Answer:Firstly it has to be in 1NF which it is,and it has been made to full functional dependency. No partial dependency.That's why it is in 2NF form.

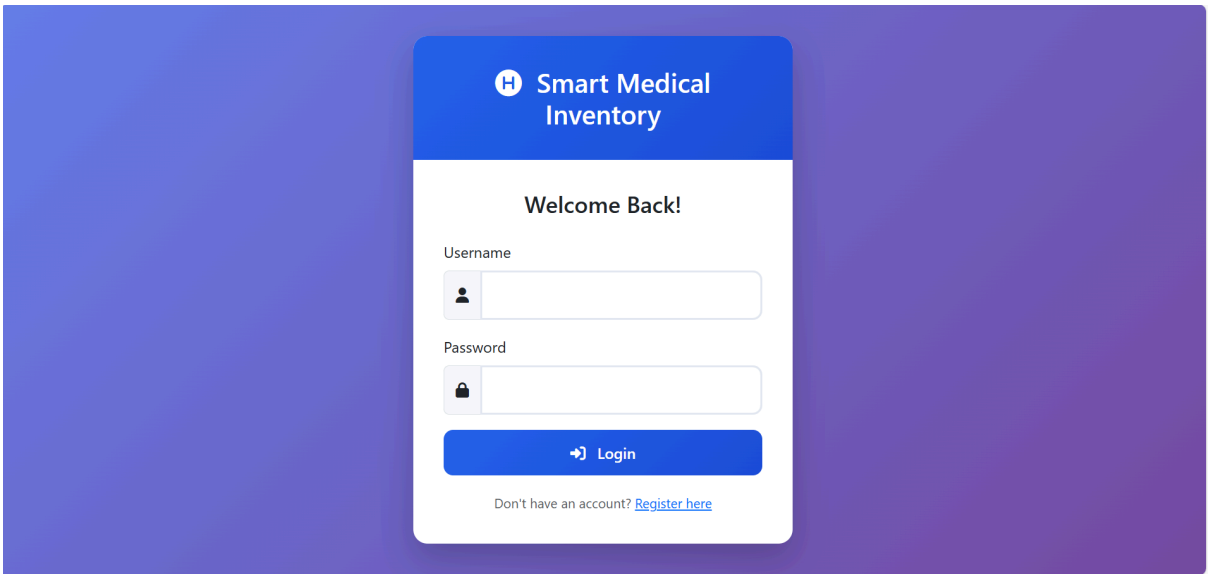
c. Answer: It has been decomposed to 3NF. It can not have transitive dependencies.

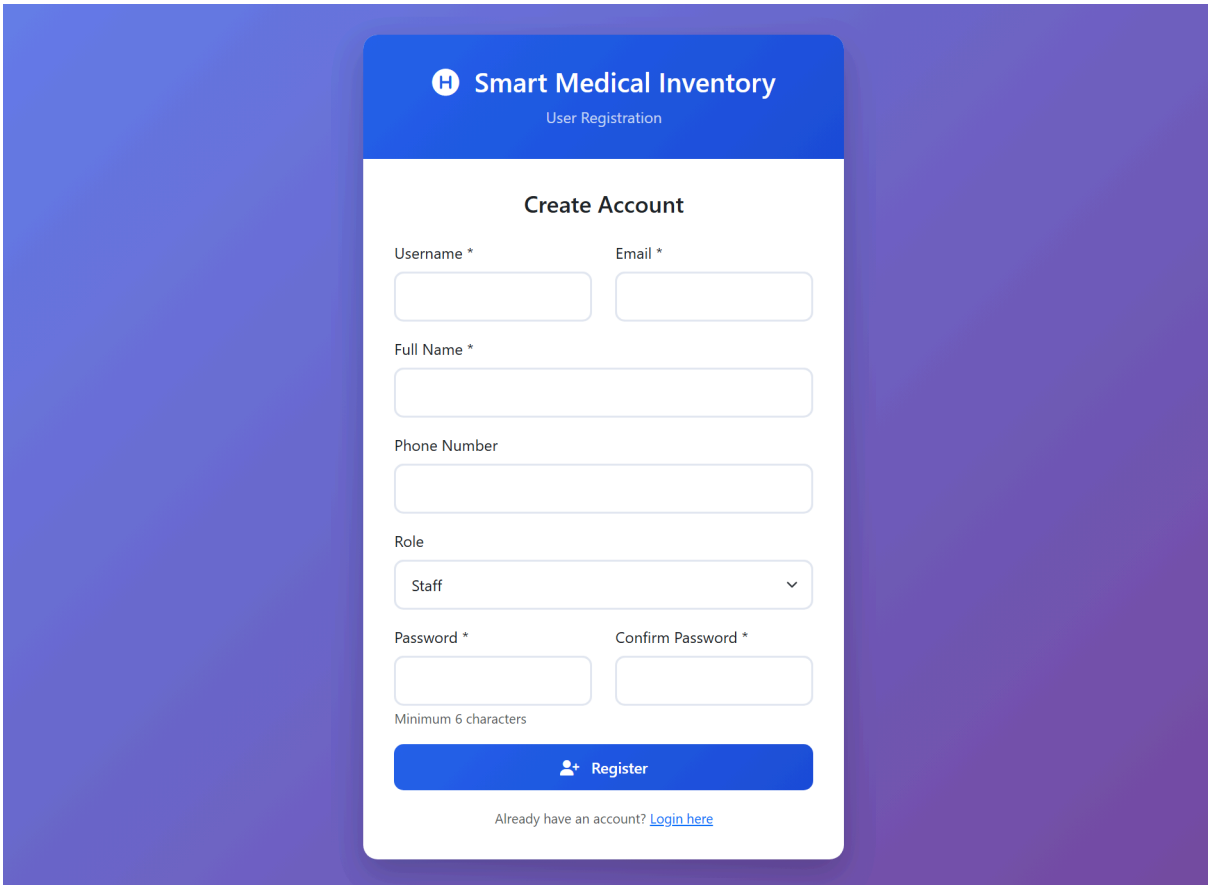
Frontend Development

Our platform leverages **Bootstrap 5.3.0, Font Awesome 6.4.0, custom CSS, and a PHP backend** to deliver a **modern, mobile-first solution** with a public site, secure authentication, role-based dashboards, and real-time content management. With **gradient design, responsive layouts, and smooth interactions**, it ensures a fast, scalable, and engaging experience for SuperAdmins, Admins, Staff, and Suppliers.

Contribution of ID : 22201484, Name : Sabah Maryam

Authentication System (login.php, register.php) - User login/registration interfaces

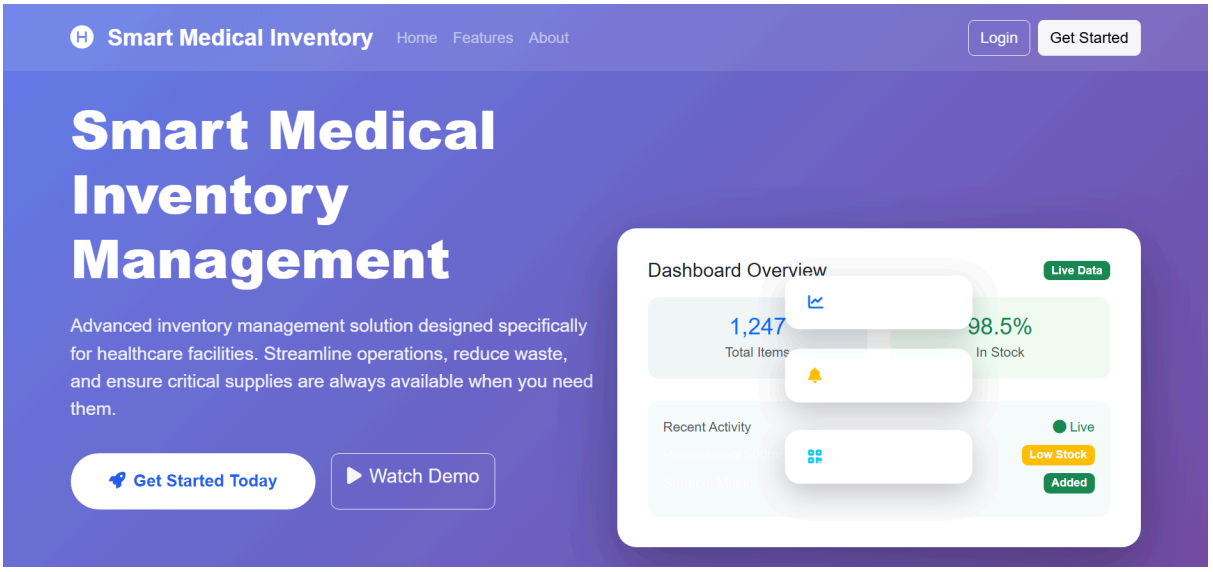


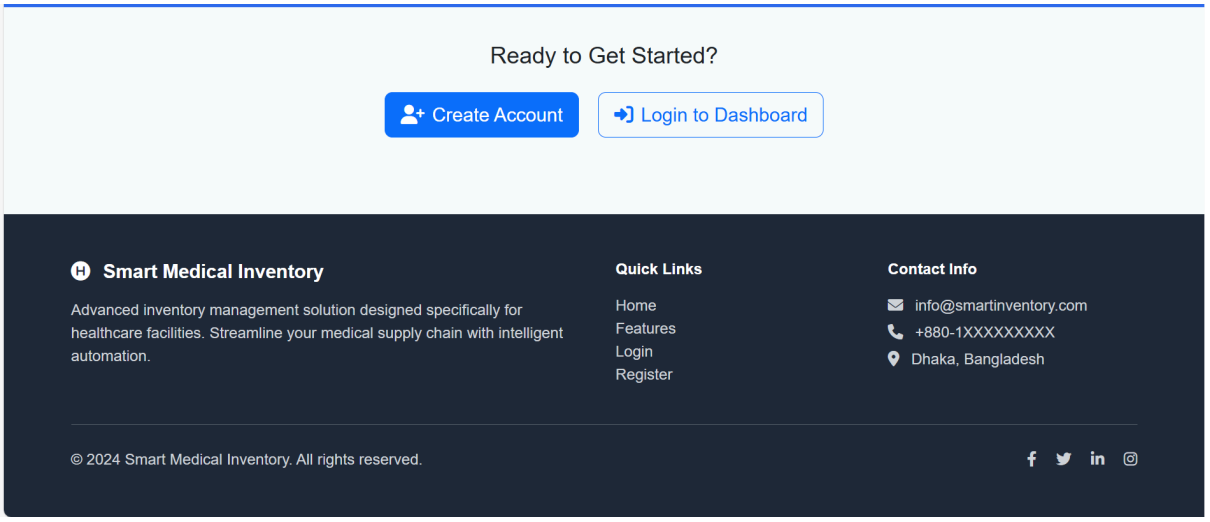


Contribution of ID : 21301563, Name : Shezan Mahmud

Hero Section CSS Framework & Styling - Bootstrap integration, custom CSS, color schemes, responsive design

UI/UX Design - Overall visual design, component styling, animations, hover effects





Contribution of ID : 21301655, Name : Izaz Mahmud Evan

Landing Page (frontend.php) - Features showcase , statistics display

Smart Inventory Tracking

Real-time monitoring of medical supplies with automated stock level alerts and expiry date tracking.

Role-Based Access

Secure user management with different access levels for administrators, staff, and suppliers.

Advanced Analytics

Comprehensive dashboard with real-time statistics and detailed reporting capabilities.

Mobile Responsive

Smart Notifications

Secure & Reliable

Trusted by Healthcare Professionals

Join thousands of healthcare facilities already using our platform

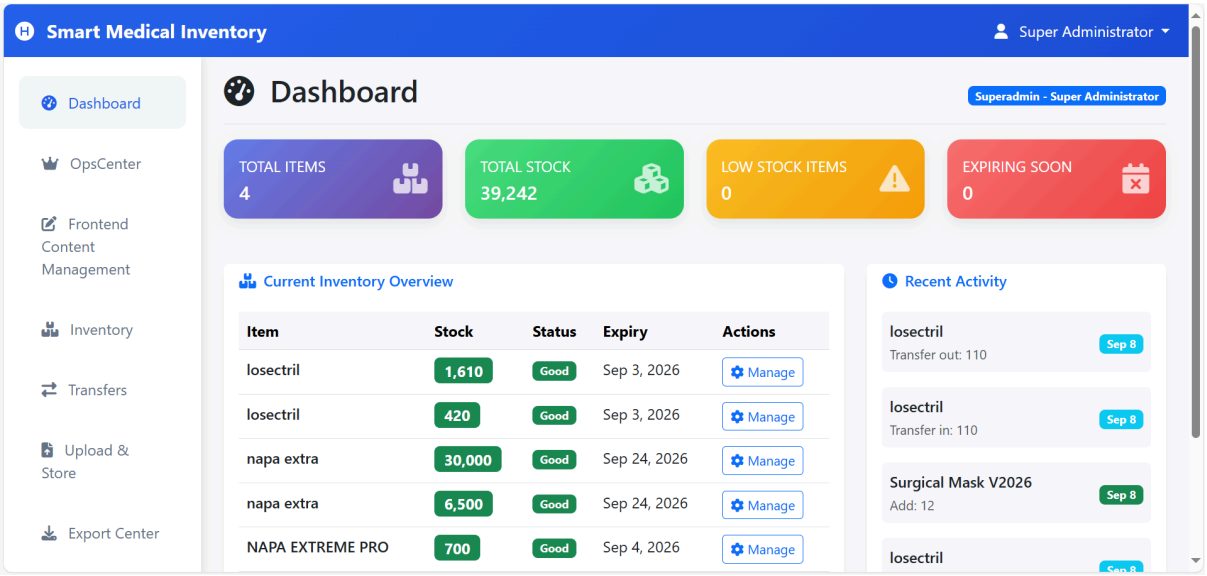
500+	1M+	99.9%	24/7
Healthcare Facilities	Items Managed Daily	System Uptime	Customer Support

Backend Development

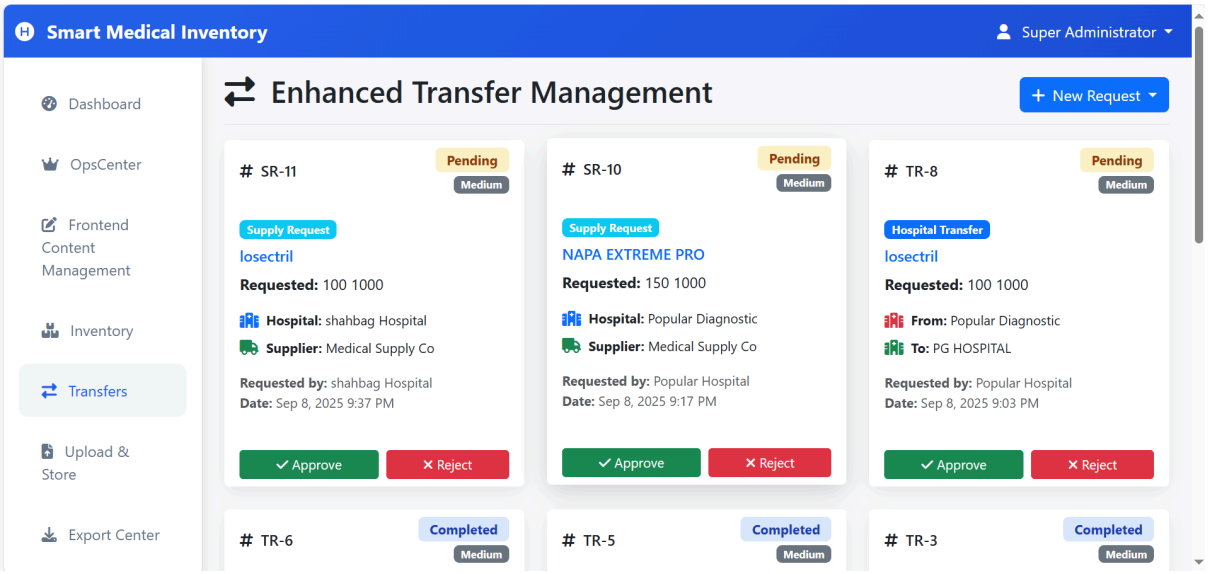
The Smart Medical Inventory Management System streamlines hospital and supplier operations with a secure, scalable backend. It supports multiple user roles with tailored permissions for seamless collaboration.The system enables real-time inventory tracking and automated reporting for smarter decisions.Built-in transfer management ensures smooth coordination across hospitals and suppliers.
Overall, it's a complete enterprise solution designed for efficiency, security, and growth.

Contribution of ID : 22201484, Name : Sabah Maryam

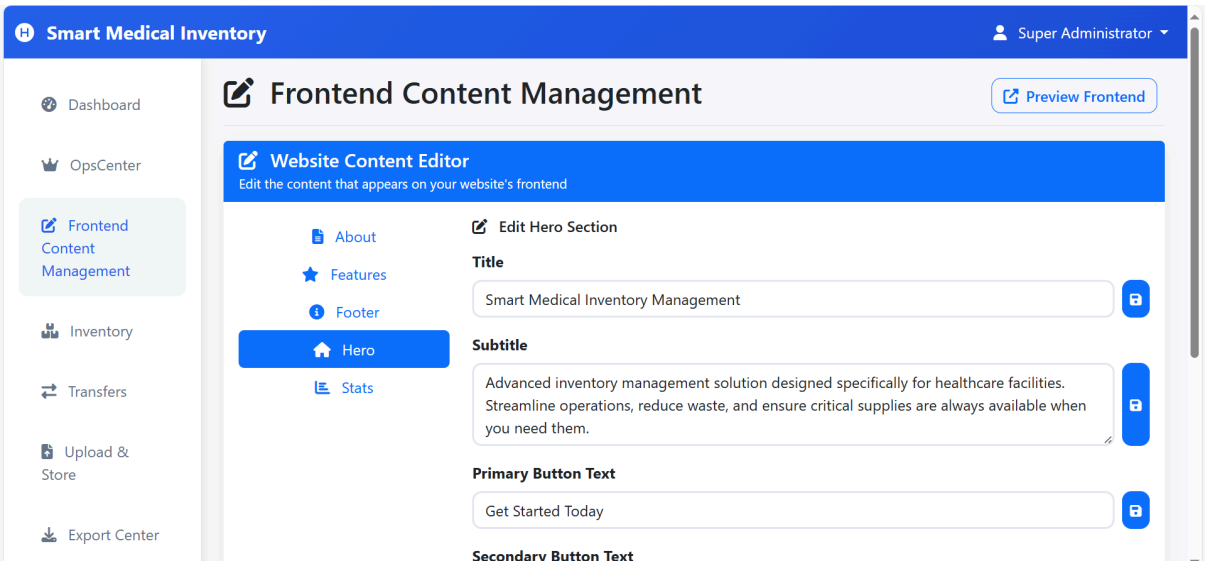
- **Dashboard** → Developed the main control panel for users to view system status, analytics, and quick actions.



- **Transfer** → Implemented hospital–supplier stock transfer functionality with approval workflows.

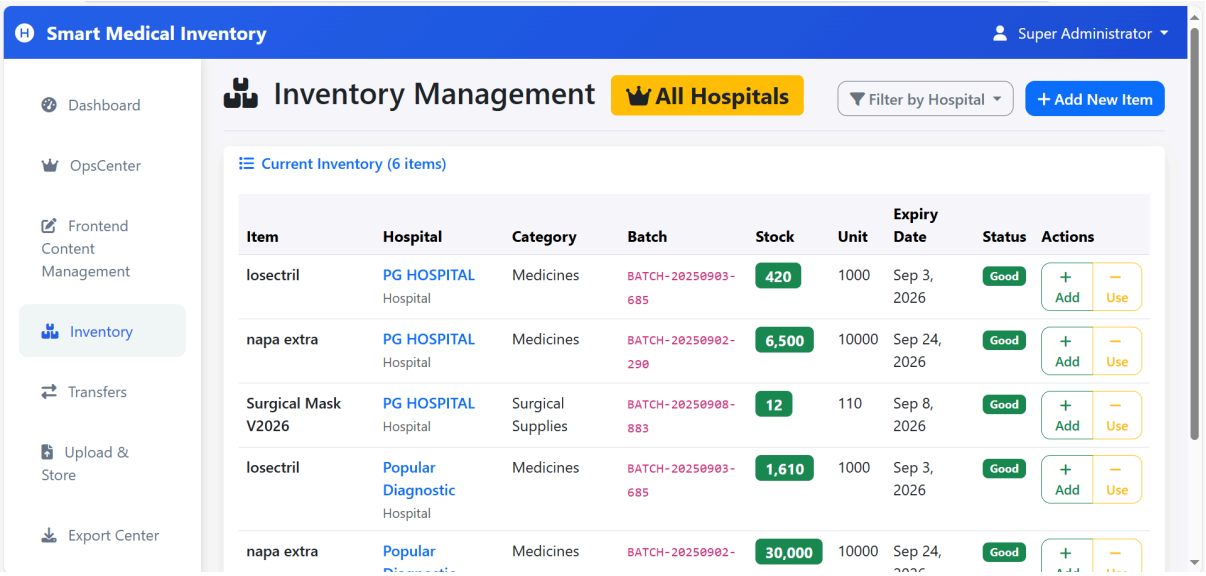


- **Frontend Content Management (Header)** → Designed and managed the website header section, ensuring easy customization.

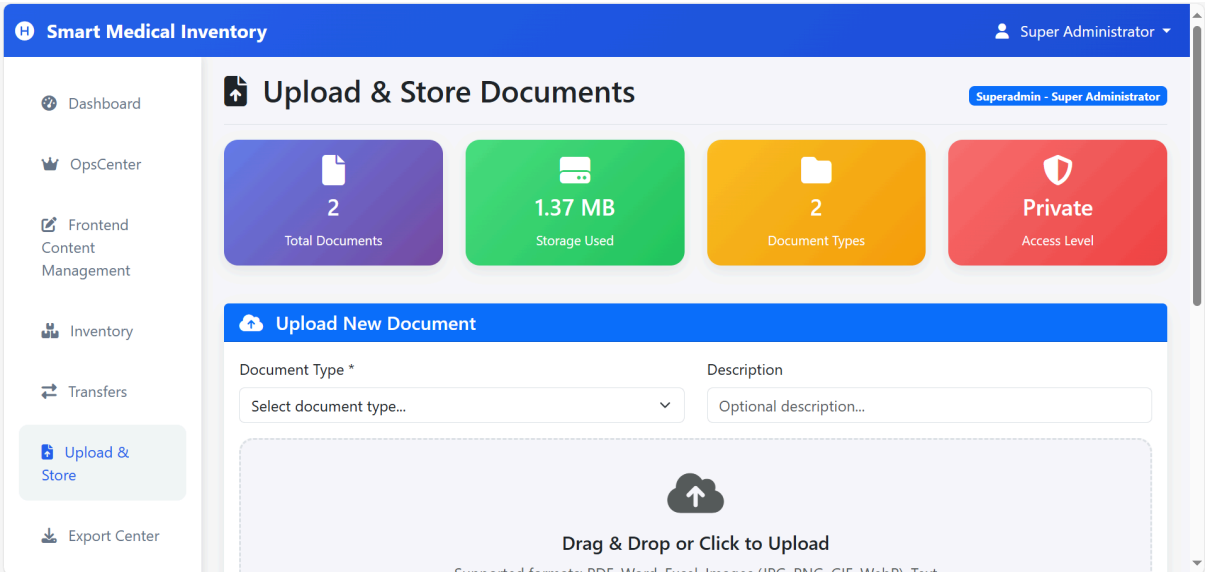


Contribution of ID : 21301563, Name : Shezan Mahmud

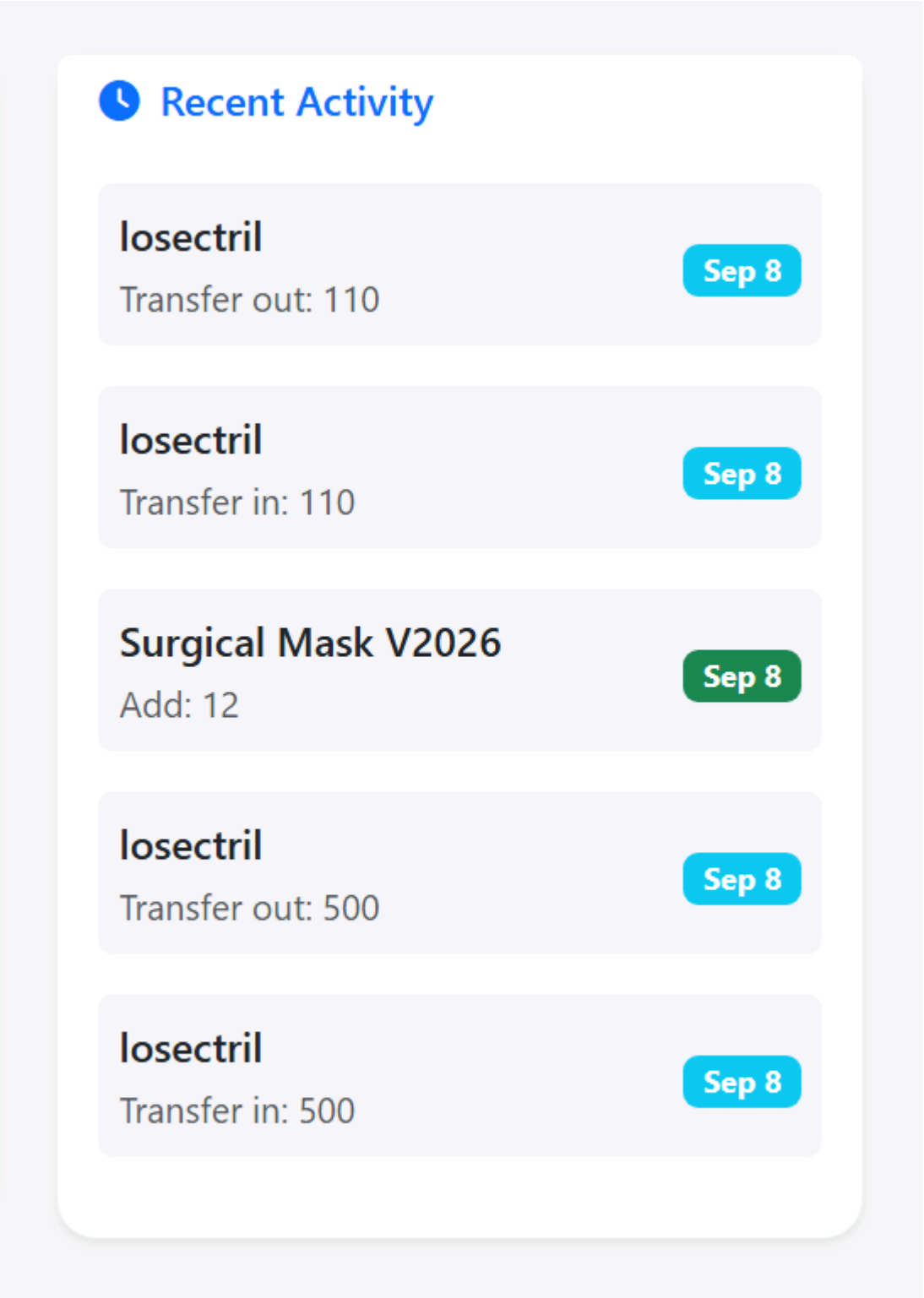
- **Inventory Management** → Built the module to track medicines, equipment, and stock levels in real-time.



- **Document Manager** → Implemented secure storage, upload, and retrieval of medical documents.

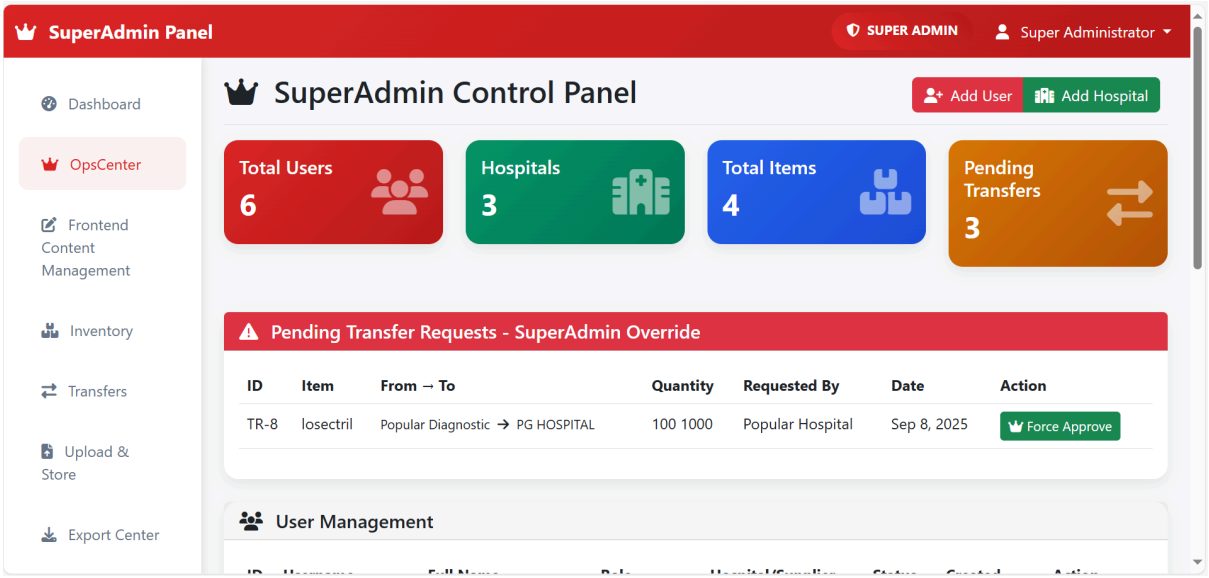


- **Shortage & Expiry Alerts**→ Automated notifications for low stock and items nearing expiration, customizable alert thresholds.

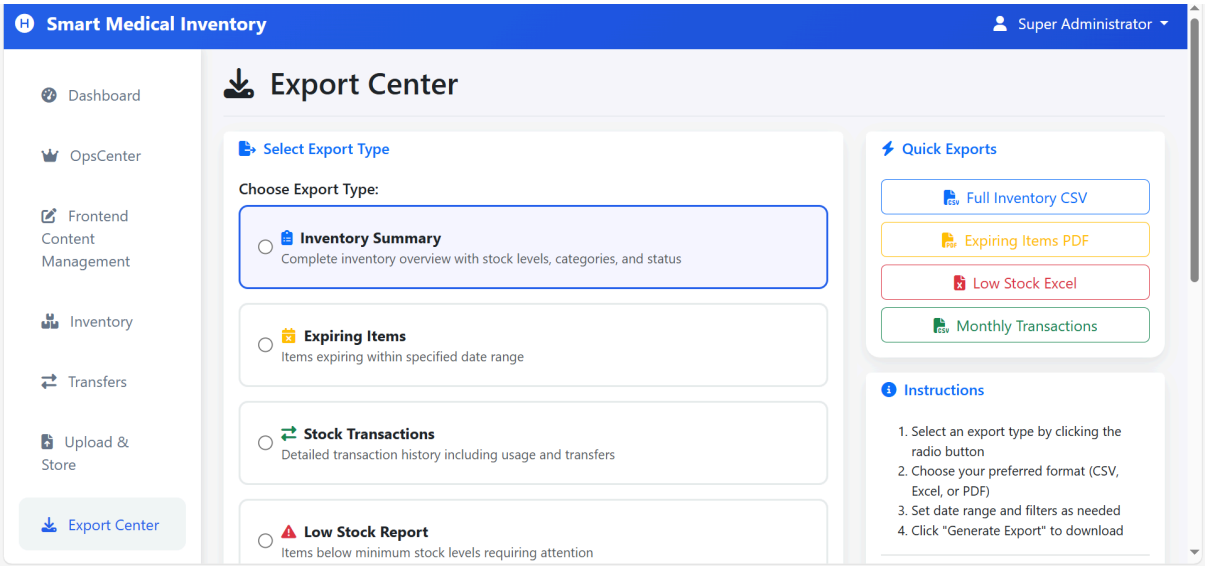


Contribution of ID : 21301655, Name : Izaz Mahmud Evan

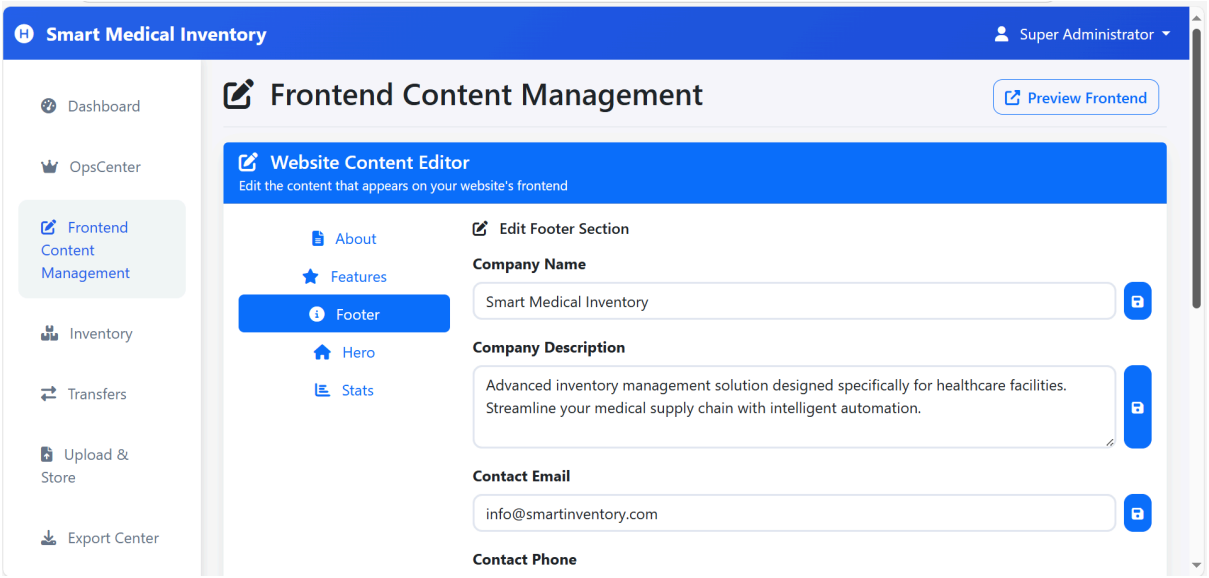
- **OpsCenter** → Created the operations management hub for overseeing hospital workflows.



- **Export Center** → Developed functionality for exporting reports and inventory data in various formats.



- **Frontend Content Management (Footer)** → Managed and customized the footer section for branding and additional information.


Smart Medical Inventory
Super Administrator
Dashboard
OpsCenter
Frontend Content Management
Inventory
Transfers
Upload & Store
Export Center
Frontend Content Management
Preview Frontend
Website Content Editor
Edit the content that appears on your website's frontend
About
Features
Footer
Hero
Stats
Edit Footer Section
Company Name
Smart Medical Inventory
Company Description
Advanced inventory management solution designed specifically for healthcare facilities. Streamline your medical supply chain with intelligent automation.
Contact Email
info@smartinventory.com
Contact Phone

Source Code Repository

https://drive.google.com/drive/folders/1r5Rj2B6eFRlDh27Dl4B_Iv9cBwwnYVZM?usp=sharing

Conclusion

The Smart Inventory Database for Medical Supplies provides an effective, technology-driven solution to overcome the common challenges of inventory mismanagement, shortages, and wastage in the healthcare sector of Bangladesh. By integrating hospitals, clinics, and suppliers into a single platform with real-time tracking, automated alerts, and collaborative dashboards, the system ensures seamless supply chain operations and timely availability of critical medical resources. Its role-based access control, detailed reporting, and supplier management features further enhance security, transparency, and accountability. Ultimately, this platform not only optimizes resource utilization but also strengthens coordination between healthcare providers and suppliers, thereby contributing to improved patient care, reduced operational bottlenecks, and a more resilient healthcare infrastructure.

References

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